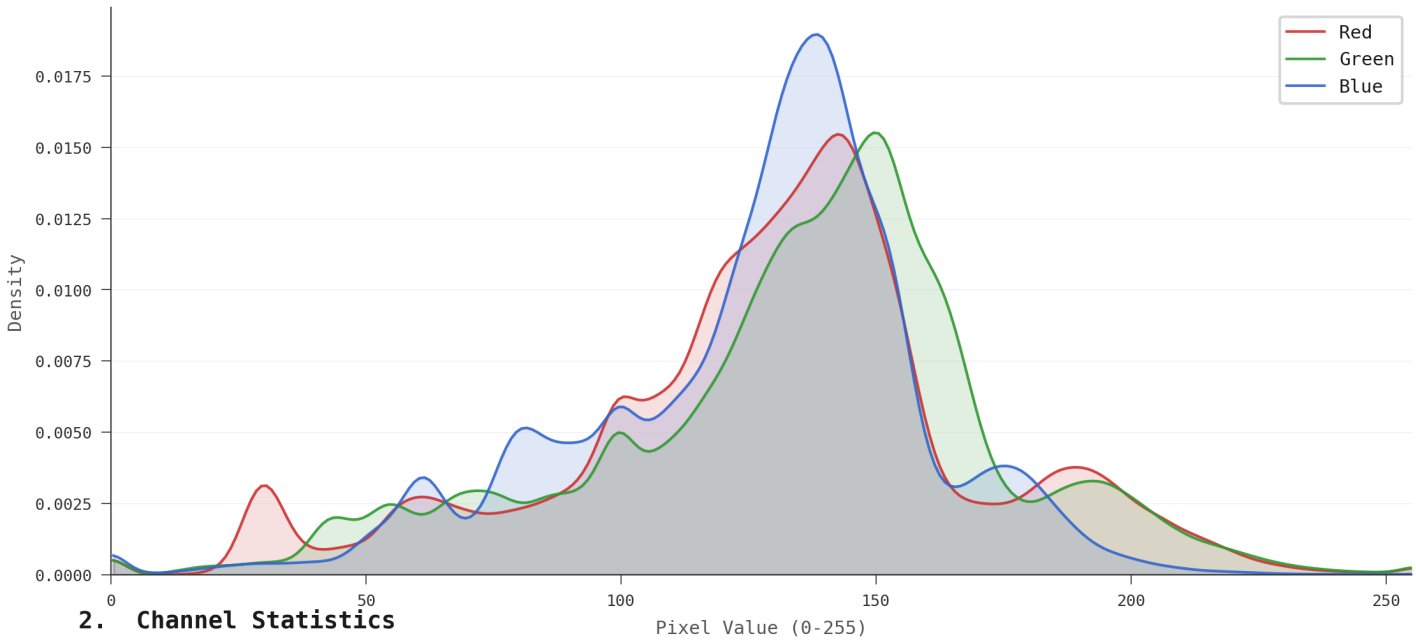
Reference Data Sheet: KODIM09

Subject: Spinnaker Sailboats Racing

512x768 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	129.2367	40.6342	1651.1352	0.5117	-0.3450	0	255
Green	134.3520	39.5654	1565.4214	0.4616	-0.4582	0	255
Blue	125.9906	33.2819	1107.6857	0.7139	-0.5961	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.9486	0.7330
G	0.9486	1.0000	0.8560
B	0.7330	0.8560	1.0000

Pairwise Pearson Correlation Coefficients

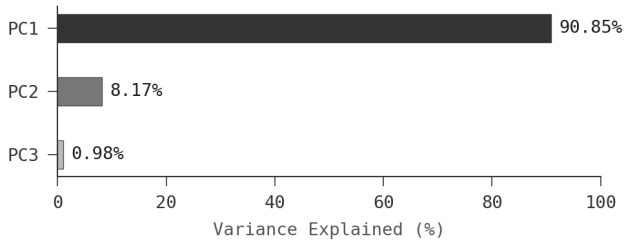
r(R-G)	=	**0.948560**
r(R-B)	=	**0.732954**
r(G-B)	=	**0.856028**
Avg |r|	=	**0.845847**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{1651.1394} & 1525.0105 & 991.2362 \\
1525.0105 & \mathbf{1565.4254} & 1127.2310 \\
991.2362 & 1127.2310 & \mathbf{1107.6885}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM09

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.6232	**-0.6258**	-0.4690	3928.7410
PC2	-0.5660	-0.0530	0.8227	353.1306
PC3	0.5398	-0.7781	0.3211	42.3817

6. Derived Metrics

Condition Number

92.70

(l1/l3) PC1/PC2: 11.13 | PC2/PC3: 8.33

Blue Channel Independence

22.0%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Green

Loading: -0.6258 | Green dominant

Classification: Weakly One-Dimensional (PC1 = 90.85%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.952365	0.966268	0.959316
Green	0.946314	0.962531	0.954422
Blue	0.934099	0.953785	0.943942

8. Average Local Variance (3x3)

Channel	Value
Red	117.5630
Green	126.2179
Blue	104.9096

9. Redundancy Profile

Classification: Weakly One-Dimensional | Condition Number: 92.70 | PC1: 90.85%

Correlation: R–G pair leads at r = 0.948560 | G–B: 0.856028 | R–B: 0.732954

Blue Channel: 22.0% independent | Substantial independence from PC1

Spatial: Avg autocorrelation: 0.952560 | Avg local variance: 116.2302

Eigenvector: Green dominant | G (-0.6258) leads | R (-0.6232), B (-0.4690)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.
[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."
[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.
[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."